

Mobile Sales Analysis Dashboard Using Power BI

1. Project Overview

Project Title

Mobile Sales Analysis Dashboard

Domain

Retail Sales Analytics

Tool Used

- Microsoft Power BI
 - Power Query
 - DAX
 - Microsoft Excel (Dataset)
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2. Business Problem

A mobile retail company generates thousands of sales transactions across multiple cities, brands, and payment methods. However, the business lacks a centralized dashboard to monitor sales performance, customer behavior, and product trends.

The objective of this project is to build an interactive Power BI dashboard that enables stakeholders to:

- Monitor overall sales performance.
 - Identify top-performing mobile brands and models.
 - Analyze customer purchasing behavior.
 - Compare payment method usage.
 - Track monthly and daily sales trends.
 - Support data-driven business decisions.
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3. Business Objectives

The dashboard aims to answer the following business questions:

- What is the total sales revenue?
- How many mobile phones were sold?
- Which brands generate the highest sales?
- Which mobile models are most popular?
- Which cities contribute the most revenue?

- What payment methods do customers prefer?
 - How do sales vary across months and weekdays?
 - What are the customer satisfaction ratings?
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4. Dataset Information

Dataset Name

Mobile Sales Transactions Dataset

Number of Columns

13

Attributes

Column	Description
Transaction ID	Unique transaction identifier
Day	Purchase day
Month	Purchase month
Year	Purchase year
Day Name	Weekday of transaction
Brand	Mobile phone brand
Units Sold	Quantity sold
Price Per Unit	Selling price
Customer Name	Customer details
Customer Age	Customer age
City	Sales location
Payment Method	Payment mode
Customer Ratings	Customer feedback
Mobile Model	Purchased mobile model

5. Data Cleaning & Preparation

Data preprocessing was completed using **Power Query**.

Cleaning Steps

- Verified data types.
- Renamed columns for consistency.
- Removed duplicate records (if any).
- Checked for blank or null values.
- Ensured numerical fields were properly formatted.
- Prepared date-related fields for time-based analysis.

6. Data Modeling

This project uses a **single-table data model**, making it straightforward to analyze sales without creating table relationships.

7. Dashboard KPIs

The dashboard provides a high-level overview of sales performance.

KPI	Value
Total Sales	769M
Total Quantity Sold	19K
Total Transactions	4K
Average Sales per Transaction	40K

8. Dashboard Components

KPI Cards

Displays overall business performance using:

- Total Sales
 - Total Quantity
 - Total Transactions
 - Average Sales
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Sales by City (Map)

Visualizes total sales geographically.

Purpose:

- Identify high-performing cities.

- Compare regional performance.
 - Detect growth opportunities.
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Monthly Sales Trend

Shows monthly quantity sold.

Business Value:

- Detect seasonality.
 - Monitor monthly demand.
 - Support inventory planning.
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Customer Ratings

Displays customer satisfaction distribution.

Business Value:

- Measure customer experience.
 - Identify service improvement opportunities.
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Sales by Payment Method

Compares payment preferences among customers.

Business Value:

- Understand payment trends.
 - Improve payment infrastructure.
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Brand Performance

Displays

- Total Sales
- Quantity Sold
- Transactions

for every mobile brand.

Business Value:

- Compare brand contribution.
- Identify leading brands.

Mobile Model Performance

Ranks mobile models by total sales.

Business Value:

- Identify best-selling products.
- Support inventory decisions.

Sales by Day Name

Shows sales distribution across weekdays.

Business Value:

- Determine peak shopping days.
- Optimize staffing and promotions.

9. Dashboard Filters

Interactive slicers allow users to analyze data by:

- Mobile Model
- Brand
- Payment Method
- Month

This enables dynamic exploration of the data from multiple perspectives.

10. Key Business Insights

Sales Performance

- The company generated **769M** in total sales.
- Approximately **19K mobile units** were sold.
- Around **4K transactions** were completed.

Customer Behavior

- Customers use all payment methods relatively evenly, indicating diverse payment preferences.
- Customer ratings are concentrated around higher scores, suggesting generally positive customer satisfaction.

Product Performance

- Certain mobile models contribute significantly more sales than others.
- Brand-wise analysis highlights clear differences in revenue contribution, helping identify top-performing manufacturers.

Geographic Analysis

- Sales are distributed across multiple cities.
- Major metropolitan areas contribute a larger share of total revenue.

Time Analysis

- Monthly sales remain relatively stable with noticeable peaks during certain months.
 - Sales vary across weekdays, helping identify the busiest shopping days.
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11. Business Recommendations

Increase Inventory

Maintain higher inventory levels for top-selling mobile models to reduce stock-out risk.

City-Based Marketing

Allocate additional marketing resources to high-performing cities while launching promotional campaigns in lower-performing regions.

Brand Partnerships

Strengthen relationships with top-performing mobile brands and negotiate exclusive offers.

Customer Experience

Continue monitoring customer ratings and investigate lower-rated transactions to improve service quality.

Payment Optimization

Promote digital payment methods through cashback or reward programs to encourage adoption.

Seasonal Promotions

Schedule promotional campaigns during months with lower sales to improve overall revenue consistency.

12. Project Outcome

The Power BI dashboard transforms raw transaction data into actionable business insights by providing a centralized, interactive reporting solution.

Decision-makers can monitor sales performance, evaluate customer behavior, analyze geographic trends, and make informed business decisions using a single dashboard.

13. Skills Demonstrated

- Data Cleaning using Power Query
 - Data Modeling
 - DAX Calculations
 - KPI Design
 - Interactive Dashboard Development
 - Data Visualization
 - Business Analysis
 - Sales Analytics
 - Retail Reporting
 - Dashboard Design Best Practices
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14. Conclusion

The **Mobile Sales Analysis Dashboard** provides a comprehensive view of sales performance across products, locations, customers, and time. By integrating key performance indicators with interactive visualizations, the dashboard enables stakeholders to identify trends, evaluate business performance, and make data-driven decisions. It serves as an effective reporting solution for monitoring sales activities and supporting strategic planning.
